MAX PLANCK INSTITUTE
FOR DYNAMICS OF COMPLEX
TECHNICAL SYSTEMS
MAGDEBURG

DMP
DATA-DRIVEN MODELING OF
COMPLEX PHYSICAL SYSTEMS

Molecular Kinetics with Koopman Operators and Random Fourier Features

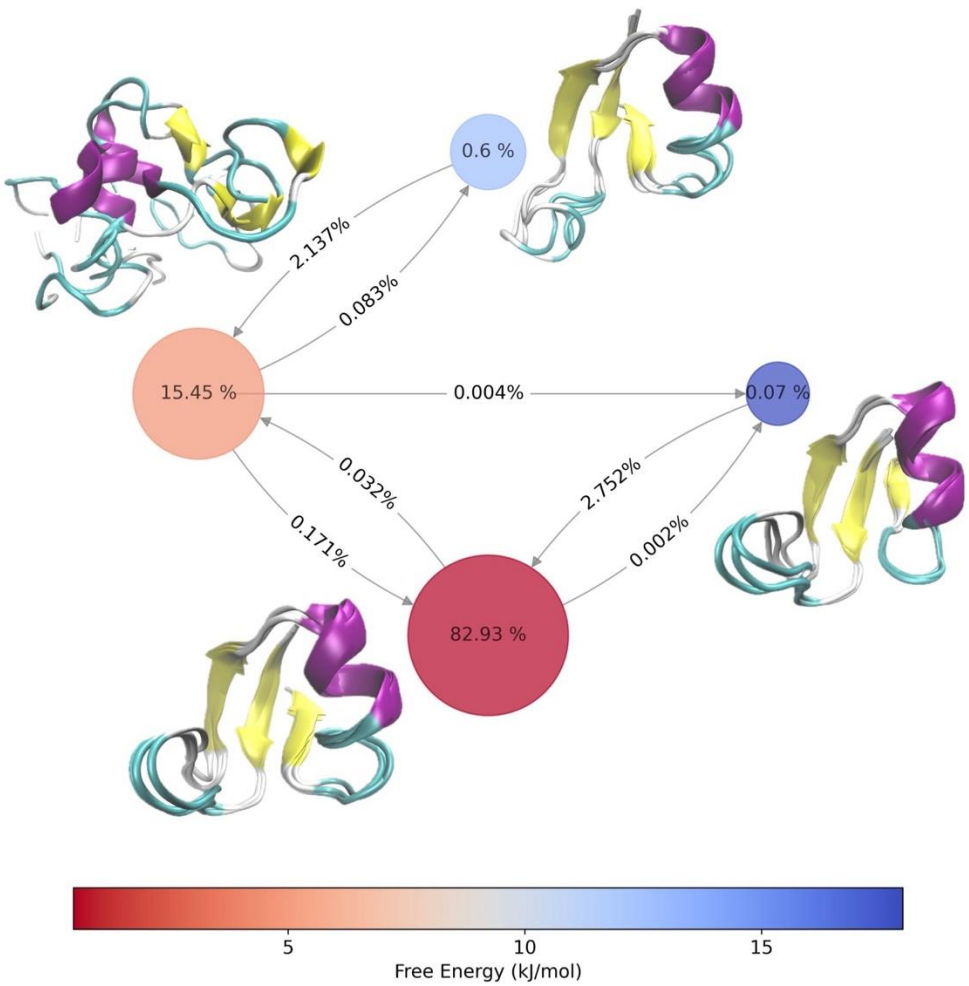
Workshop on Intelligent Computational Methods in Molecular Dynamics
Shanghai Jiao-Tong University

Feliks Nüske, MPI Magdeburg

Modeling Dynamical Processes

Molecular Dynamics provides atomistic-level insights into dynamical processes:

- protein folding
- ligand binding
- protein-protein association
- …



Challenges (for this talk):

- Determination of meta-stable states and transition rates
- Determine effective models on coarse-grained spaces

Contents

I. Modeling Slow Transitions with Koopman Operators
II. Generator Approximation and Reduced Models

Koopman Operator

Conditional Expectation of Observables

$$\mathcal{K}^t f(x) = \mathbb{E}\left[f(X_t) \,|\, X_0 = x\right]$$

Convergence to equilibrium:

$$\mathcal{K}^t f(x) = \sum_{j=0}^{\infty} e^{-\kappa_j t} c_f \psi_j(x) = \mathbb{E}^{\pi}[f] + \sum_{j=1}^{\infty} e^{-\kappa_j t} c_f \psi_j(x)$$

Slowly decaying modes and relaxation timescales:

$$\kappa_j \ll 1, 0 \leq j \leq N \qquad t_j = \frac{1}{\kappa_j} = -\frac{t}{\log(\lambda_j(t))}$$

Meta-stable states and transition rates are discoverable from Koopman spectral decomposition.

Koopman, PNAS (1931)
Mezic, Nonlinear Dyn. (2005)
Klus, **FN,** Wu, Koltai, Kevrekidis, Schütte, and Noé, JNLS (2018)

Variational Theorem

Eigenvalues maximize equilibrium auto-correlations:

$$\sum_{j=0}^{N} \lambda_j(t) = \max \sum_{j=0}^{N} \mathbb{E}^{\pi}\left[f_j(X_0)\, f_j(X_t)\right]$$

$$\delta_{ij} = \mathbb{E}^{\pi}\left[f_i(X_0) f_i(X_0)\right]$$

Matrix Approximation from finite basis set:

$$\psi_j \approx \sum_{l=1}^{n} \mathbf{a}_{jl}\phi_l, \qquad \mathbf{C}^t\mathbf{a}_l = \lambda_l\, \mathbf{C}^0\mathbf{a}_l$$

with correlation matrices:

$$\mathbf{C}^t = \mathbb{E}^{\pi}\left[\phi(X_0) \otimes \phi(X_t)\right], \qquad \mathbf{C}^0 = \mathbb{E}^{\pi}\left[\phi(X_0) \otimes \phi(X_0)\right].$$

Matrix approximation of Koopman operator (EDMD):

$$\mathbf{K}^t = (\mathbf{C}^0)^{-1}\, \mathbf{C}^t$$

Noé and **FN**, SIAM MMS (2013)
FN, Keller, Pérez-Hernández, Mey, and Noé, JCTC (2014)
Klus, **FN,** Wu, Koltai, Kevrekidis, Schütte, and Noé, JNLS (2018)

Variational Theorem

Data-driven version of Variational theorem:

$$\sum_{j=0}^{N} \lambda_j(t) \approx \max \frac{1}{m} \sum_{j=0}^{N} \sum_{s=1}^{m} f_j(x_{s\Delta_t})\, f_j(x_{s\Delta_t+t})$$

$$\delta_{ij} = \frac{1}{m} \sum_{s=1}^{m} f_j(x_{s\Delta_t})\, f_j(x_{s\Delta_t})$$

Approximation of correlation matrices:

$$\mathbf{C}^t \approx \frac{1}{m} \sum_{s=1}^{m} \phi(x_{s\Delta_t}) \otimes \phi(x_{s\Delta_t+t}), \qquad \mathbf{C}^0 \approx \frac{1}{m} \sum_{s=1}^{m} \phi(x_{s\Delta_t}) \otimes \phi(x_{s\Delta_t+t})$$

Noé and **FN**, SIAM MMS (2013)
FN, Keller, Pérez-Hernández, Mey, and Noé, JCTC (2014)
(See also) Wu and Noé, JNLS (2020)

Random Feature Basis

Many different ways to choose the basis set:

- Markov State Models: indicator functions of discrete sets
- Time-lagged independent component analysis (TICA): basic molecular descriptors (distances, torsions, …)
- Machine-learned non-linear transformations, e.g. VAMPnets
- …

Random Fourier Features (RFF)

with random frequencies drawn from spectral measure ρ, and uniformly random phase shifts.

$$\phi_k(x) = \sin(\omega_k^\top x + b_k)$$

Approximates kernel-based method with kernel k determined by spectral measure.

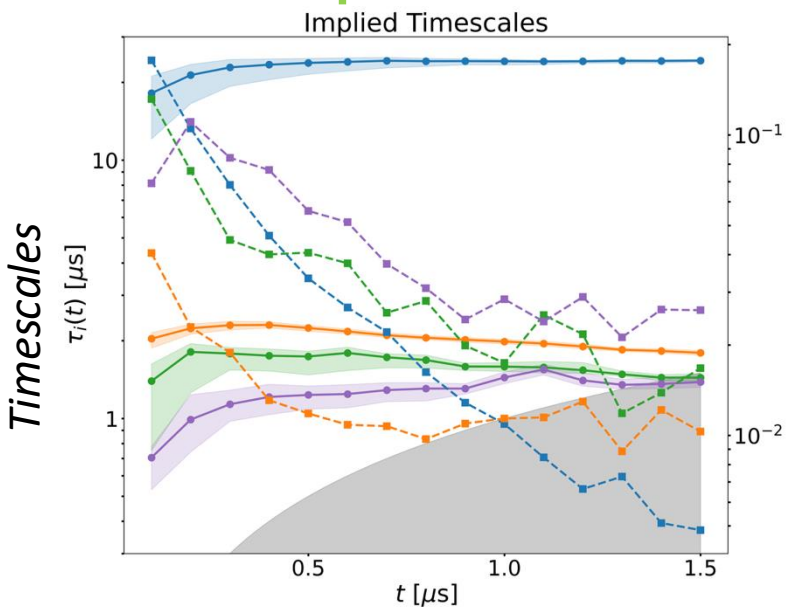
Rahimi and Recht, Neurips (2007)
FN and Klus, JCP (2023)

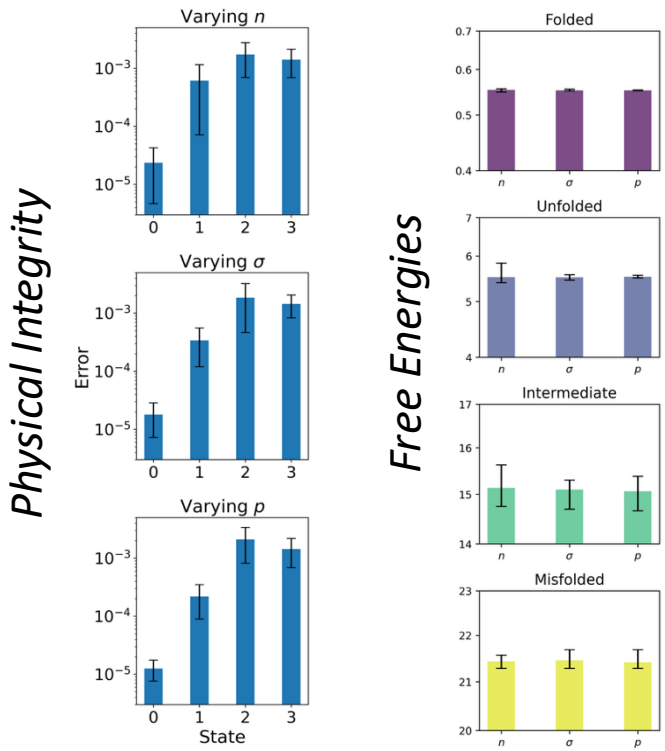
(See also)
Prinz et al, JCP (2011)
Pérez-Hernández et al, JCP (2013)
Mardt et al, Nat. Comm. (2018)

Kinetics from Random Features

Variational Approximation with Random Features:

Robust under parameter selection:





Sprink, Zhu, Mey, and **FN,** to appear (2025)

Contents

I. Modeling Slow Transitions with Koopman Operators
II. Generator Approximation and Reduced Models

Generator Approximation

Koopman Ops form a semigroup, with generator:

$$\mathcal{L}f = \lim_{h \to 0} \frac{1}{h} \left[\mathcal{K}^h - \mathrm{Id} \right] f$$

Generator is typically a differential operator. For Langevin dynamics:

$$\mathcal{L}f = \left[-\nabla V \cdot \nabla + k_B T \, \Delta \right] f.$$

Generator models are useful for:

1. **Kinetic models from Boltzmann samples alone**
2. **Kinetic properties of coarse-grained models**
3. **Interpolation across temperature space**

Generator Approximation

Variational theorem:

$$-\sum_{j=0}^{N} \kappa_j = \max \sum_{j=0}^{N} \mathbb{E}^{\pi}\left[f_j(X_0)\,\mathcal{L}f_j(X_0)\right] = \langle f_j,\, \mathcal{L}f_j\rangle_{\pi}$$

$$\delta_{ij} = \mathbb{E}^{\pi}\left[f_i(X_0) f_j(X_0)\right] = \langle f_i, f_j\rangle_{\pi}$$

Matrix Approximation (gEDMD):

$$\mathbf{L} = (\mathbf{C}^0)^{-1}\,\mathbf{C}^{\mathrm{gen}}, \qquad \mathbf{C}^{\mathrm{gen}} = \langle \phi,\, \mathcal{L}\phi\rangle_{\pi}, \qquad \mathbf{C}^0 = \langle \phi,\, \phi\rangle_{\pi}$$

- Does not require transition pairs, only Boltzmann samples

Zhang, Li and Schütte (2022)
Klus, **FN**, Peitz, Niemann, Schütte, and Clementi, Physica D (2020)

CG Generator

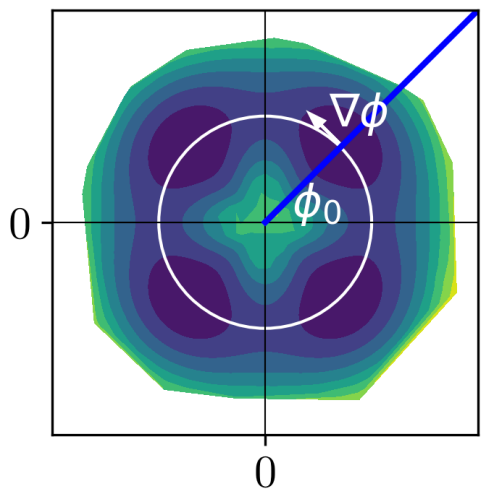
Consider a CG map ξ and the Zwanzig projection:

$$\mathcal{P}f(z) = \mathbb{E}^{x\sim\pi}\left[f(x)\,|\,\xi(x) = z\right]$$

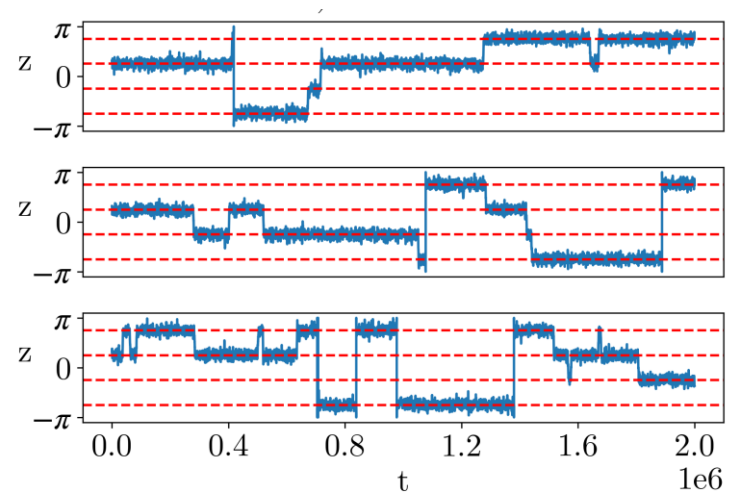
CG generator: Markovian term in Mori-Zwanzig decomposition:

$$\mathcal{L}^{\xi} = \mathcal{P}\mathcal{L}\mathcal{P}$$

CG Map



CG Dynamics



CG Generator

Relation between generators:

$$\langle \mathcal{L}^\xi f, g\rangle_\nu = \langle \mathcal{P}\mathcal{L}\mathcal{P} f, g\rangle_\nu = \langle \mathcal{L}(f\circ\xi), g\circ\xi\rangle_\pi.$$

- **gEDMD models for the CG generator can be learned from full space data.**

$$\mathbf{L}^\xi = (\mathbf{C}^0_\xi)^{-1}\,\mathbf{C}^{\mathrm{gen}}_\xi, \qquad \mathbf{C}^{\mathrm{gen}}_\xi = \langle \phi, \mathcal{L}^\xi\phi\rangle_\nu, \qquad \mathbf{C}^0_\xi = \langle \phi, \phi\rangle_\nu$$

- **Eigenvalues and timescales of the CG dynamics can be calculated**

$$\mathbf{C}^{\mathrm{gen}}_\xi \mathbf{a}_l = \kappa_l\,\mathbf{C}^0_\xi \mathbf{a}_l.$$

- **Parameters (effective diffusion) of the CG dynamics can be learned:**

$$a^\xi = \operatorname{argmin} \frac{1}{m}\sum_{i=1}^{m} \|a(z_i) - a^\xi_{\mathrm{loc}}(x_i)\|_F^2,$$

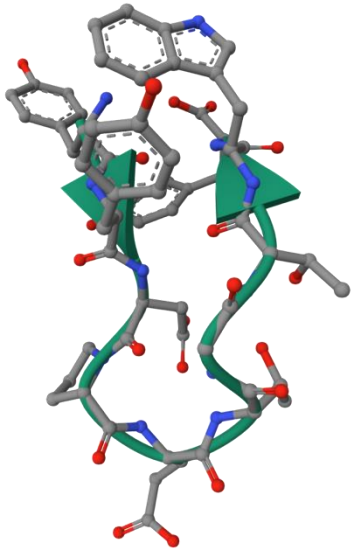
Zhang, Hartmann and Schütte, Faraday Disc. (2016)
Nateghi and **FN**, JCTC (2025)

Learning CG Dynamics

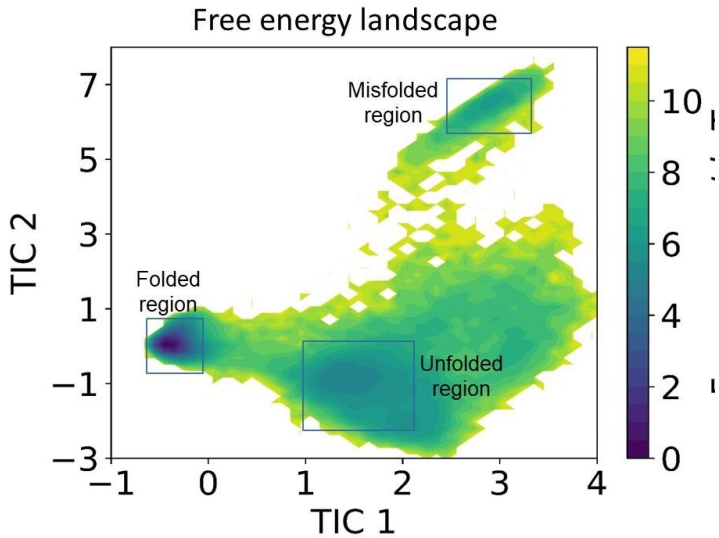
Chignolin Example:

- projection into 2d TICA space
- learn gEDMD model with RFF basis
- learn effective diffusion with RFF basis

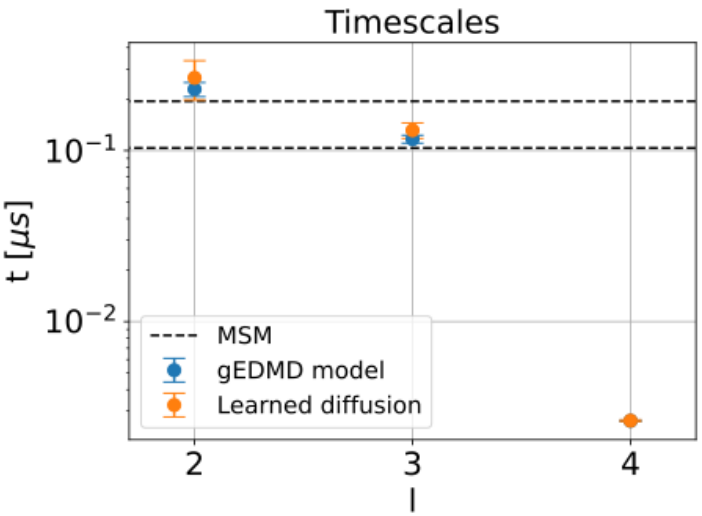
Full Space



CG Space



Kinetic Properties

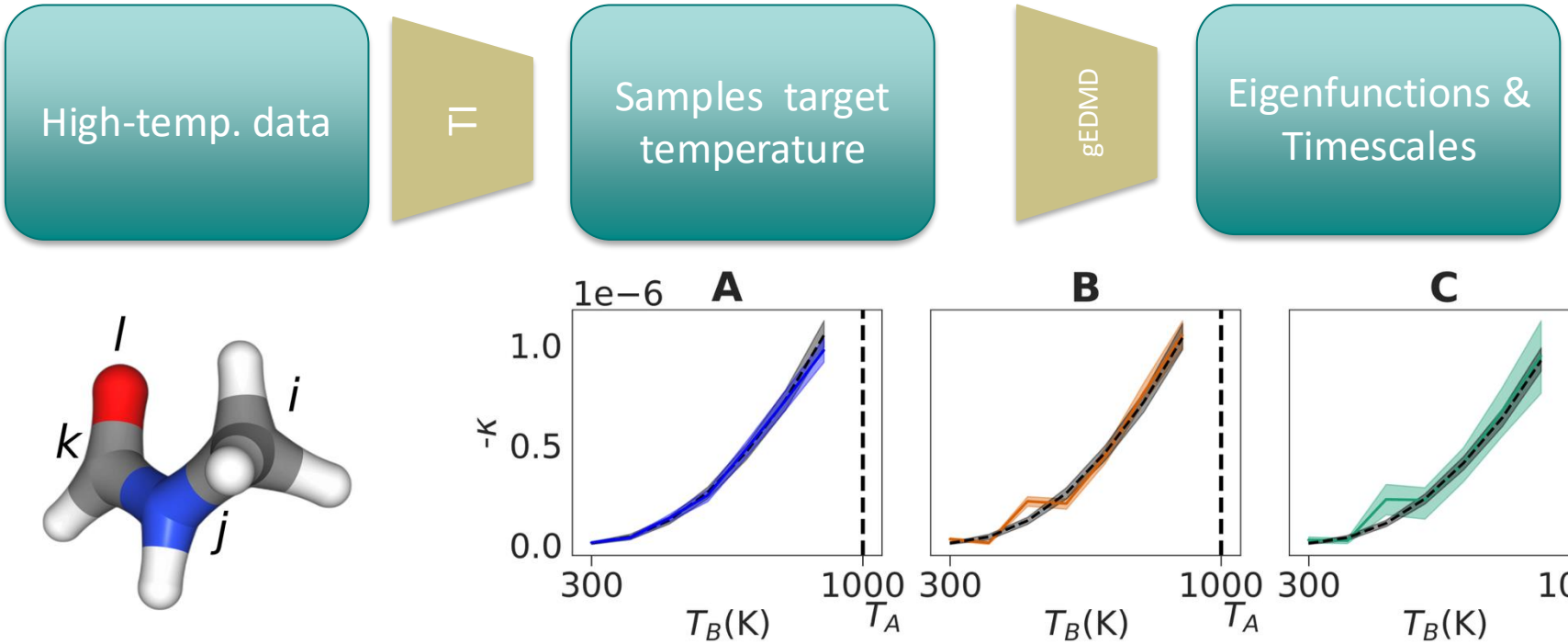


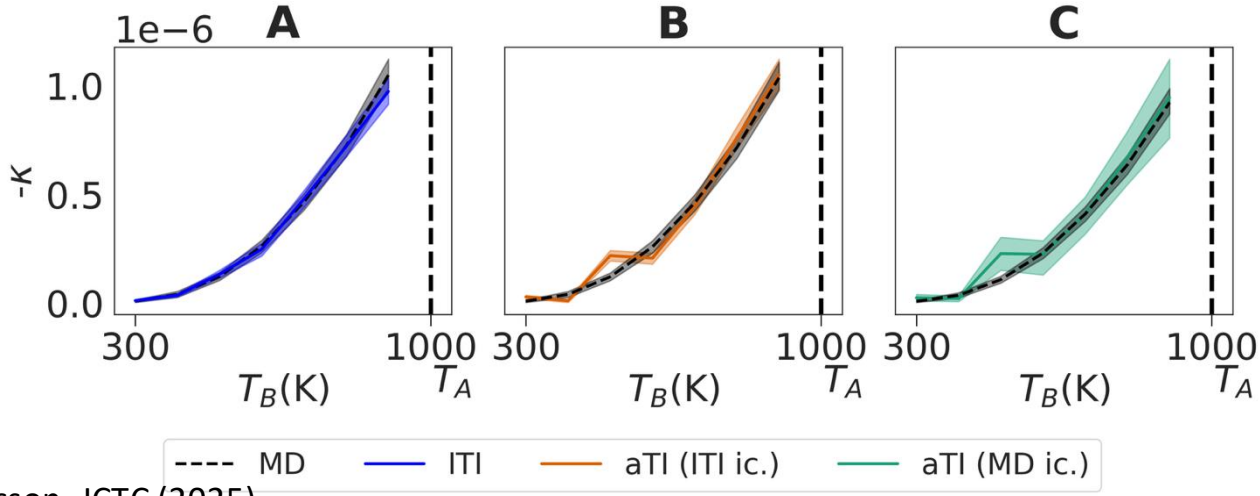
Nateghi and **FN**, JCTC (2025)

Thermodynamic Interpolation

gEDMD only requires Boltzmann samples -> can directly learn kinetic properties from generative models.

- Use thermodynamic interpolation (TI) to train generative model based on high-temperature data, and extrapolate to target temperature.





Moqvist, Chen, Schreiner, **FN**, and Olsson, JCTC (2025)

Thanks for your Attention!